

Cong Yang

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Education

2023–2026	Ph.D. in Computer Science • School of Computer Science, Wuhan University • Advisor: Lefei Zhang (zhanglefei@whu.edu.cn) • GPA: 3.71/4.0	Wuhan, China
2020–2023	M.S. in Geological Resources and Geological Engineering • School of Geophysics and Geomatics, China University of Geosciences • Advisor: Yanni Dong (dongyanni@whu.edu.cn) • GPA: 3.62/4.0	Wuhan, China
2016–2020	B.S. in Earth Information Science and Technology • School of Geophysics and Geomatics, China University of Geosciences • GPA: 3.35/4.0	Wuhan, China

Research Interests

Multimodal Learning, Image Captioning, Post-Training

Research Experience

2026–Present	Associative and Diversity-Aware Reinforcement Learning for Multimodal Reasoning <i>Doctoral Researcher, Advisor – Prof. Lefei Zhang</i> • Built the Associative CoT-9K dataset by restructuring linear chains of thought into dependency-linked question–answer subproblem chains, enabling structured and traceable multimodal reasoning. • Implemented a two-stage post-training pipeline comprising supervised fine-tuning on Associative CoT-9K followed by group relative policy optimization (GRPO). • Introduced a semantic-diversity regularizer using Qwen3-Embedding-4B to penalize homogeneous reasoning trajectories and extend optimization from final-answer accuracy to reasoning-path quality. • Evaluated Qwen2.5-VL-7B and Qwen3-VL-8B across multimodal reasoning benchmarks, developing end-to-end experience in reinforcement-learning post-training, ablation design, trajectory analysis, training-cost profiling, and first-author manuscript preparation (M1).
2023–2025	Vision–Language Learning for Fine-Grained Image Captioning and Change Understanding <i>Doctoral Researcher, Advisor – Prof. Lefei Zhang</i> • Developed vision–language pre-training and instruction-tuning methods for image–text alignment, fine-grained captioning, and change understanding. • Introduced BITA, a two-stage framework that combines multi-scale frequency features, contrastive learning, and prefix causal language modeling to improve vision–language alignment. • Curated 61,783 multimodal instruction pairs for fine-grained captioning, instruction tuning, and evaluation.

	• Designed MGIMM and KCFI to integrate regional and global feature fusion, key-change perception, and multi-task optimization for detailed description and change caption generation. • Validated these methods on challenging remote sensing datasets, resulting in three first-author articles: J4, J5, and J6.
2021–2022	Metric Learning and Domain Adaptation under Limited Annotations Graduate Researcher, Advisor – Prof. Yanni Dong • Developed DMLOHM, combining discriminative metric-space learning with online hard-triplet mining to improve representation learning under limited annotations. • Achieved gains of 0.13%–4.03% with only 85 labeled samples per class, building experience in sparse-label experimental design, metric-learning evaluation, and difficult-sample analysis. • Designed ADA-DDA to assign attention-based sample weights and dynamically balance marginal and conditional distribution alignment across domains, improving accuracy by 5.36% over the next-best method. • Applied these methods to hyperspectral and cross-domain remote sensing classification, contributing to J1, J2, J3, and a granted Chinese invention patent (CN114202697B).

Publication

Journal Articles

- 2025 J6 **Enhancing Perception of Key Changes in Remote Sensing Image Change Captioning**
Cong Yang, Zuchao Li, Hongzan Jiao, Zhi Gao, Lefei Zhang
IEEE Transactions on Image Processing
DOI: [10.1109/TIP.2025.3589096](https://doi.org/10.1109/TIP.2025.3589096)
- 2024 J5 **MGIMM: Multi-Granularity Instruction Multimodal Model for Attribute-Guided Remote Sensing Image Detailed Description**
Cong Yang, Zuchao Li, Lefei Zhang
IEEE Transactions on Geoscience and Remote Sensing
DOI: [10.1109/TGRS.2024.3497976](https://doi.org/10.1109/TGRS.2024.3497976)
- J4 **Bootstrapping Interactive Image–Text Alignment for Remote Sensing Image Captioning**
Cong Yang, Zuchao Li, Lefei Zhang
IEEE Transactions on Geoscience and Remote Sensing
DOI: [10.1109/TGRS.2024.3359316](https://doi.org/10.1109/TGRS.2024.3359316)
- 2022 J3 **Attention-Based Dynamic Alignment and Dynamic Distribution Adaptation for Remote Sensing Cross-Domain Scene Classification**
Cong Yang, Yanni Dong, Bo Du, Liangpei Zhang
IEEE Transactions on Geoscience and Remote Sensing
DOI: [10.1109/TGRS.2022.3225589](https://doi.org/10.1109/TGRS.2022.3225589)
- J2 **Joint Distance Transfer Metric Learning for Remote-Sensing Image Classification**
Yanni Dong, Tianyang Liang, **Cong Yang**, Hui Luo, Yuxiang Zhang
IEEE Geoscience and Remote Sensing Letters
DOI: [10.1109/LGRS.2022.3149141](https://doi.org/10.1109/LGRS.2022.3149141)
- 2021 J1 **Deep Metric Learning with Online Hard Mining for Hyperspectral Classification**
Yanni Dong, **Cong Yang**, Yuxiang Zhang
Remote Sensing
DOI: [10.3390/rs13071368](https://doi.org/10.3390/rs13071368)

Preprints

- 2026 M1 **AD-GRPO: Associative Diversity-Aware GRPO for Multimodal Reasoning**

Service

Peer Reviewing

Journals

- *IEEE Transactions on Geoscience and Remote Sensing* (TGRS)
- *IEEE Transactions on Image Processing* (TIP)
- *IEEE Transactions on Circuits and Systems for Video Technology* (TCSVT)

Awards

- 2023 Outstanding Graduate of China University of Geosciences
2021 China National Scholarship

Skills

Research Expertise	Remote sensing image understanding and generation; Multimodal learning; Model post-training
Programming & Computing	Python, MATLAB, C/C++; PyTorch, Hugging Face Transformers, DeepSpeed; Slurm, distributed multi-GPU computing, Git
Languages & Writing	Mandarin Chinese (native); English (professional working proficiency; experienced in academic writing and manuscript preparation)